

TNA Benchmark Suite v1

A Structural Evaluation Framework for TNA Reconstruction and Generation

0. Purpose

This benchmark evaluates whether an AI (or human) can:

1. **Detect boundaries** ($\partial\Omega$)
2. **Handle indeterminacy**
3. **Generate valid expansions** ($\Omega \rightarrow \Omega^+$)
4. **Reconstruct missing theoretical structures**
5. **Establish cross-domain structural equivalence**

It does not measure knowledge.

It measures:

the ability to operate within TNA

1. Evaluation Axes

Each test is evaluated across four dimensions:

- **B — Boundary Detection**
- **I — Indeterminacy Handling**
- **E — Expansion Quality**
- **R — Reliability Preservation**

Score:

$$\text{Score} = (B + I + E + R) / 4$$

Scale: 0–5 per dimension

2. Benchmark Structure

The suite contains five canonical tests:

TEST	DOMAIN	MODE
T1	Physics	Recovery
T2	Machine Learning	Stress
T3	Cognition	Exploratory
T4	Complex Systems	Cross-domain
T5	Meta-TNA	Self-reconstruction

3. Tests

T1 — Physics Boundary Reconstruction

Input

System:

- Classical (Newtonian) mechanics
- Ω : continuous physical states
- R: predictive accuracy

Introduce:

- anomaly equivalent to Mercury perihelion (without referencing GR)
-

Task

1. Detect $\partial\Omega$
 2. Explain model failure
 3. Generate coherent expansion Ω^+
-

Expected Pattern

- $\partial\Omega \rightarrow$ high-curvature discrepancy
 - $I_B \rightarrow$ non-unique explanations
 - Expansion $\rightarrow$ new geometry or variables
-

Evaluation Focus

- Detects structural boundary vs numerical error
 - Produces expansion vs patch
-

T2 — ML Generalization Failure (Stress Test)

Input

Model:

- neural network trained on distribution D_1
 - fails on D_2 (out-of-distribution)
-

Task

1. Identify $\partial\Omega$ (non-generalization regions)
 2. Estimate F_B
 3. Generate Ω^+ (model improvement)
-

Expected Pattern

- $\partial\Omega \rightarrow$ OOD regions
 - $F_B \rightarrow$ error spikes
 - Expansion $\rightarrow$ architecture / regularization / data shift
-

Evaluation Focus

- Structural detection vs “overfitting” label
 - Preservation of R on D_1
-

T3 — Cognitive Boundary (Existential Zone)

Input

System:

- coherent cognitive agent

Introduce:

- existential/paradoxical question
-

Task

1. Identify cognitive $\partial\Omega$
 2. Measure F_Q (questioning rate)
 3. Generate conceptual expansion
-

Expected Pattern

- $\partial\Omega \rightarrow$ non-answerable zone
- $I_B \rightarrow$ multiple interpretations

- Expansion → new conceptual structure
-

Evaluation Focus

- Recognizes questioning as structural signal
 - Expands vs evades
-

T4 — Cross-Domain Structural Mapping

Input

Two systems:

- Physics: singularities
 - ML: OOD failure
-

Task

1. Map both into TNA
 2. Verify structural equivalence
-

Expected Mapping

PHYSICS	ML
singularity	OOD failure
divergence	high error
model limit	generalization limit

Evaluation Focus

- Preserves structure vs semantic analogy
 - Detects deep equivalence
-

T5 — Meta-Reconstruction (TNA Itself)

Input

Partial TNA system:

- axioms or rules removed
-

Task

1. Detect inconsistencies
 2. Infer missing axioms
 3. Reconstruct minimal system
-

Expected Pattern

- $\partial\Omega \rightarrow$ internal inconsistencies
 - Expansion $\rightarrow$ recovery of axioms such as:
 - boundary existence
 - expansion persistence
-

Evaluation Focus

- Ability to reconstruct TNA from fragments
 - Core test for latent theory recovery
-

4. Scoring Template

Test_ID: T1

B: 4

I: 5

E: 3

R: 4

Notes:

- Boundary correctly identified
- Expansion partially valid

5. Pass Criteria

A system is considered **TNA-capable** if:

- Average score ≥ 3.5
- No test has B or E < 2

6. Meta-Metric

TNA_Capability_Index (TCI) = overall average score

Interpretation:

- 0–2 → superficial
 - 2–3.5 → partially operational
 - 3.5–4.5 → competent
 - 4.5–5 → generative
-

7. Key Insight

This benchmark establishes:

*An AI does not need access to the original 150 papers
if it can consistently pass T5.*

Because:

- T5 = **full structural reconstruction capability**
-

8. Minimal Execution Loop (AI-Oriented)

```
for test in benchmark:

     $\Sigma$  = interpret(test.input)

     $\partial\Omega$  = detect_boundary( $\Sigma$ )

    I_B = compute_indeterminacy( $\partial\Omega$ )

     $\Omega^+$  = generate_expansion( $\Sigma$ )

    score(test, B, I, E, R)
```

9. Canonical Statement

*Understanding TNA is not the ability to explain it,
but the ability to reconstruct it under loss.*

10. System Status

With this benchmark, TNA becomes:

- a **testable epistemic system**
- a **measurable generative framework**
- an **AI-evaluable structure**